VSL-B1: information and departure-time examples

13 October 2026 — original conditional deductions, not an adopted model

1. Combined continuing mass need not determine adjacent receiving

Consider equal-length local sections in three lanes: ending E, adjacent continuing A, and farther continuing F. Length is ℓ [m]; densities k_E, k_A, k_F are in vehicle-equivalents/m. For this diagnostic only, use the standard receiving relation

$$R(k) = \min\{C,\ w(k_j - k)\},$$

where C is a rate [vehicle-equivalents/s], $w > 0$ is a backward-wave speed [m/s], and k_j is jam density. This is a source-compatible receiving ingredient, not a selected VSL or lane-transfer update.

Choose $0 \le k_L < k_H \le k_j$ with $R(k_H) < R(k_L)$. The two snapshots

$$(k_E, k_A, k_F) = (k_E, k_H, k_L), \qquad (k_E, k_A, k_F) = (k_E, k_L, k_H)$$

have the same ending mass ℓk_E and combined continuing mass $\ell(k_H + k_L)$, but different receiving opportunity in the adjacent lane. With identical positive merge requests and an allocation in which that receiving difference binds, completed transfer can differ.

Conclusion. No function of ending and combined continuing mass alone can recover adjacent receiving for every snapshot in this set. This does not establish a reachable equal-full-history pair, a different optimal VSL, or necessity of a full lane profile. A restricted domain or compact receiving statistic may remove the ambiguity.

Initial symmetry is a separate issue. Let a, f be adjacent and farther continuing masses and let $h > 0$ be a transfer rate from E to A. If the two continuing lanes have equal net longitudinal contributions b at a comparison instant and no compensating redistribution, then

$$\dot{a} = b + h, \qquad \dot{f} = b, \qquad \frac{d}{dt}(a - f) = h > 0.$$

Thus $a = f$ initially does not imply continued equality. A dynamically valid grouped model needs an explicit equivalence condition. These equations are local conservation diagnostics, not proposed traffic evolution laws.

Primary receiving lineage: Carey–Balijepalli–Watling, Sections 3–5.
institutional manuscript; FB0030, equations 13–15, archived in the review package.

2. Equal final work can give different mean completion times

This example examines the exit interface only. All class demand is exposed at time zero. Initial continuing mass is $c_0 = 1$ and merge mass is $e_0 = 1$ [vehicle-equivalents]. Ordinary service costs one resource unit per unit mass; merge service costs $\alpha = 2$. Constant receiving work rate is $\mu = 1$ [resource units/s]. Mass is continuous; no integer vehicle process is being introduced.

Every work-conserving allocation uses total work

$$W = c_0 + \alpha e_0 = 3, \qquad T = W/\mu = 3 \text{ s}.$$

Let $N(t)$ be total mass not yet exited. Complete population time is $F = \int_0^T N(t)\,dt$ [vehicle-equivalent seconds]; mean time is $\bar{T} = F/(c_0 + e_0)$.

Continuing first. Ordinary traffic clears during the first second; merge traffic then clears over two seconds:

$$N_C(t) = \begin{cases} 2 - t, & 0 \le t \le 1, \\ (3-t)/2, & 1 \le t \le 3. \end{cases} \qquad F_C = \frac{5}{2}, \quad \bar{T}_C = \frac{5}{4} \text{ s}.$$

Merge first. Merge traffic clears during the first two seconds; ordinary traffic then clears over one second:

$$N_E(t) = \begin{cases} 2 - t/2, & 0 \le t \le 2, \\ 3 - t, & 2 \le t \le 3. \end{cases} \qquad F_E = \frac{7}{2}, \quad \bar{T}_E = \frac{7}{4} \text{ s}.$$

Proportional co-clearance. S13's conditional proportional resource projection preserves the class ratio, giving

$$N_P(t) = 2\left(1 - \frac{t}{3}\right), \quad F_P = 3, \quad \bar{T}_P = \frac{3}{2} \text{ s} = T/2.$$

Allocation	Final work	Final clearance [s]	Mean time [s]
Continuing first	3	3	1.25
Merge first	3	3	1.75
Proportional	3	3	1.50

Conclusion. Final work or clearance time alone does not determine the complete-time objective under general priorities. This is a hand-derived counterexample, not a result attributed to a source experiment. Source priority alternatives motivate checking allocation; they do not select these schedules as the proposed freeway model.

3. What these examples do and do not qualify

The examples establish two interface risks: loss of consequential receiver information and loss of consequential departure timing. Neither selects a replacement lane-transfer model, validates its common-speed response, or proves an information-to-optimal-action split.

For the fixed finite cohort with mass M, count the mass $N_{\mathrm{sys}}(t)$ still at sources or in the modeled system. With a common decision/accounting origin, complete exit at $T(v)$ gives

$$F(v) = \int_0^{T(v)} N_{\mathrm{sys}}(t; v)\, dt, \qquad \bar{T}(v) = F(v)/M.$$

If admitted trips have later release times, subtract their fixed release-time prefix in interpreting trip elapsed times. It is action-independent; it may not be confused with omitted action-dependent source waiting. No new demand release regime is selected here. The objective meaning remains the accepted complete population-time meaning even if its reduced formula changes.

Ideal travel is conditional. L/v is a traversal time only when the declared common advice v actually determines the relevant progress over length L. Binding receiving can invalidate this ideal mapping. The review does not supply an alternative speed/fundamental-diagram law.

MP6 is a joint qualification. One admitted regime must support the same geometry, finite cohort, advice response, receiving and grouping, early/late cost accounting, lane-end boundary and departures throughout its declared action range. Source-compatible ingredients and correct algebra alone do not establish that regime, calibration, or useful external gradients.

Minimal-state and comparison limits. S13's autonomous initial-ending- mass scalar remains exactly sufficient under S13. The receiver diagnostic does not invalidate that result; it concerns a different state domain. A new candidate must compare against a compact predictor given the same admissible initial information. Native reachable history matching and independent perturbation usefulness are still unproved.

Current gate. VSL-B1-G1 asks whether qualification must preserve the restricted sequential scalar/batch reduction (B1-R), or may propose the smallest local receiving and within-cohort timing amendment (B1-L). No source law or new state is adopted by this note. No simulation or parameter sweep was run.

Companion files in the review package:
`vsl_b1_lane_change_model_and_gate.md`
`vsl_b1_source_and_interface_audit.md`.